

Opening-angle resolution of the MX17 pair spectrometer — algorithm survey and hard limits

Dylan Neff — 2026-06-12 | drafted by Claude (Fable 5), reviewed by D. Neff **Data:** 10⁷ at-rest X17+IPC pair events, STEP small-capsule geometry (pairs_v2 , 2026-06-12) **Scope:** Micromegas (MM) tracking only — calorimetry assumed saturated/unusable for the high-energy measurement.

Part I — Executive summary

The number: the best opening-angle resolution any reconstruction can reach with the current hardware is $\sigma_{68}(\Delta\theta) \approx 14.5^\circ$ over the full accepted X17 spectrum, $\approx 11.5\text{--}12.5^\circ$ for the most symmetric pairs — and that already assumes the vertex is known *exactly* (oracle). The realistic versions (target-centre vertex constraint, PCA track fit) sit only 0.5–1.5° above the oracle. **All seven reconstruction methods tried land within 2° of each other.** There is no algorithmic headroom left worth chasing.

Why: the resolution is set by multiple scattering in the **capsule wall** (0.6 mm Al + 0.9 mm CFRP = **79 %** of the 1.26 % X_0 upstream budget), ~11 mm from the vertex. A scatter that close to the origin rotates the whole visible track; every downstream measurement — including a chord drawn from the true vertex — retains $\approx 96\%$ of the scattering angle (see Part II §5). No tracking upgrade, vertex constraint, or fit cleverness can undo it.

The vertex trick (your specific question): yes, it's implemented (nomline = target centre → first MM hit), and with the new small capsule it is already **within 0.5° of the perfect-vertex limit**. The small target closed that gap: assuming the capsule centre costs only ~3° per particle *in quadrature* against an 8–10° MS floor. Two-track vertexing can't beat it — individual tracks point back to the vertex with a median miss of 1.0–3.7 cm, larger than the 1 cm bore radius, so "it came from the capsule" is already the strongest vertex statement available.

What it does to the peak: the at-rest X17 truth distribution is a sharp shoulder at $\theta_{\min} \approx 109^\circ$ (fig 4). After smearing, it becomes a ~40°-FWHM bump with $\frac{1}{4}$ the peak height — sitting on the IPC continuum at trigger-level S/B ≈ 0.09 with ~84 double-triggered X17 per 30-day run. Your instinct is right: **a cut-and-count peak will not be visible.**

What can still recover the measurement (in order of leverage):

1. **Fit, don't cut.** Treat the θ spectrum as a template fit: smeared-signal shape (known from this 10M-event response) + IPC continuum shape. Resolution then enters as dilution of statistical power, not as a cliff. This is the same logic as every low-resolution bump hunt and it is *quantifiable now* with the Stage-4 pile-up sampler machinery. → recommended next step.
2. **MM-internal quality selection.** The 30–40° tails come from tracks below ~3 MeV, which are directionless after the wall. Without calorimetry these can still be tagged inside the MM (track straightness χ^2 , drift-gas dE/dx, short range). Removing them buys the 15.5° → ~12.5° gain that the truth-min-KE study demonstrates. Needs a dedicated study (current study used truth KE).
3. **Bias correction.** +3.5° constant bias at high KE, much larger at low KE — free to remove, included automatically in a response-matrix fit.

4. **Hardware** (only if 1–3 are not enough): the Al barrel alone is 53 % of the budget. An all-composite vessel would scale $\sigma \propto \sqrt{(x/X_0)} \rightarrow \approx \times 0.65$, i.e. $\sigma_{68} \approx 10^\circ$. Still a diluted peak (blue curve in fig 4) — hardware alone does not rescue cut-and-count either.

Verdict: the angular resolution is hardware-frozen at $\sigma_{68} \approx 12\text{--}15^\circ$; the recovery path is statistical (template fit + quality selection), and the go/no-go question becomes "what significance does a template fit reach with ~ 84 signal events on the realistic IPC + wall-background continuum?" — that is the natural follow-on study.

Part II — Details

1. Data set and geometry

All numbers come from the `pairs_v2` campaign: 5.0 M X17 ($m = 16.8$ MeV) + 5.0 M IPC events, generated **at rest** ($E^* = 20.58$ MeV, no neutron-energy boost) with vertices sampled uniformly in the He-3 gas, run through the full Geant4 geometry. This is the **new STEP capsule** ("smaller target"):

- He-3 gas bore: capsule shape, $r = 10$ mm, 40 mm cylinder + hemispherical end caps (60 mm total on-axis), 500 bar ($\rho = 62.7$ mg/cm³)
- Vessel: 0.6 mm Al barrel (5 mm dome on axis) + 0.9 mm CFRP wrap
- MM front face at 250 mm from origin, 4 arms; 30 mm ArIso drift gap
- Analysis: `scripts/analyze_pairs.py` $\rightarrow$ `analysis/pairs_v2/pair_analysis_v2.pdf` (17 sections) and `analysis/pairs_v2/geant4_response.json` (response tables)

Everything below uses MM information only. The trigger-scintillator method is included once, as a fallback comparison.

2. Definitions

- ψ — per-particle space angle between an estimated direction and the truth direction *at the vertex*. This is the per-track error; two tracks combine (roughly in quadrature, one plane projection each) into the pair error.
- $\Delta\theta = \theta_{\text{reco}} - \theta_{\text{truth}}$ — pair-level opening-angle residual.
- σ_{68} — half the 16–84 percentile span of $\Delta\theta$ (robust width; the tails are non-Gaussian, RMS is $\sim 25\%$ larger).

3. The estimator catalogue and what each one tests

Per-particle direction estimators (`PSI_ESTIMATORS` in `analyze_pairs.py`):

name	direction taken as	what it isolates	median ψ , all KE	KE > 8 MeV
first	true momentum at first DriftGas hit	upstream MS only — what a <i>perfect</i> tracker measuring direction at the MM entrance would get	10.4°	7.9°
last	momentum at last DriftGas hit	adds drift-gas MS	(panel only)	—

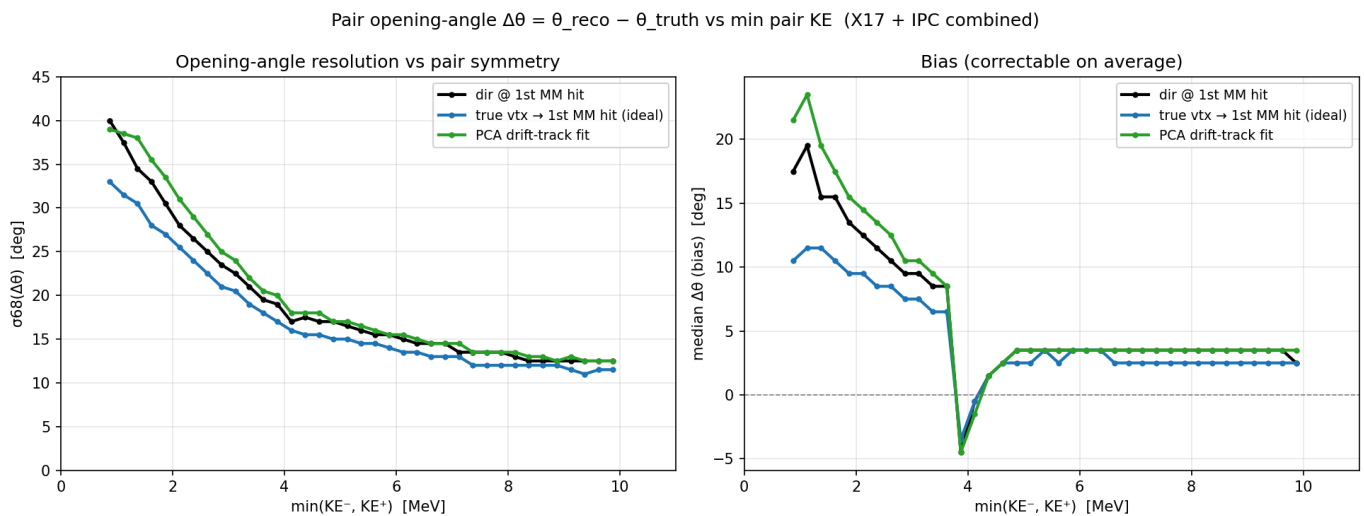
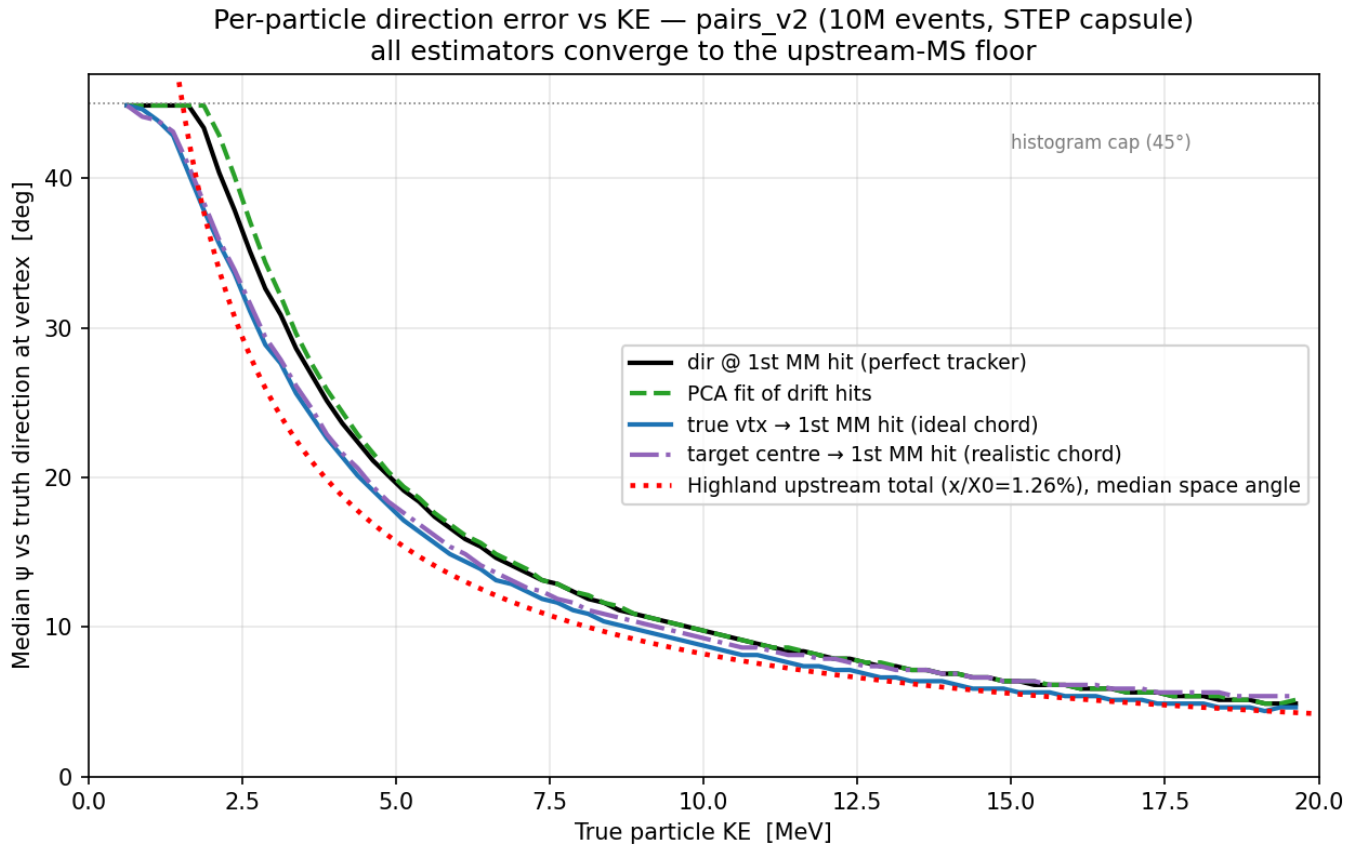
name	direction taken as	what it isolates	median ψ , all KE	KE > 8 MeV
drift	first vs last hit direction	drift-gas MS alone (small)	(panel only)	—
fit	PCA straight line through true drift-hit positions	realistic track-fit algorithm	10.4°	7.9°
fit_vs_first	PCA fit vs first-hit direction	fit-method error alone	(panel only)	—
(fits , pair level)	PCA after 0.5 mm Gaussian hit smear	digitization/cluster resolution	—	—
vline	chord: true vertex → first MM hit	oracle vertex constraint (truth not available in data)	9.4°	7.1°
vcen	chord: true vertex → edep-weighted drift centroid	centroid vs first-hit choice	9.4°	7.1°
nomline	chord: target centre → first MM hit	<i>realistic</i> vertex constraint — uses only the known capsule position	9.9°	7.9°
scint	chord: vertex → trigger-scint hit ($\sigma = 3$ cm)	scintillator-only fallback	11.4°	9.4°

Pair-level opening-angle methods and results (σ_{68} of $\Delta\theta$, full accepted sample):

method	X17	IPC	comment
first (dir @ 1st MM hit)	15.5°	15.0°	the "perfect tracker" reference
fit (PCA drift track)	16.0°	15.5°	drift-gas MS $\approx$ cancels fit averaging — no gain
fits (+0.5 mm hit smear)	16.5°	15.5°	detector resolution costs $\leq 0.5^\circ$ — MM spatial resolution is not a limiting factor
vline (true vtx chord)	14.5°	13.0°	oracle floor
vcen (true vtx → centroid)	14.5°	13.5°	same as vline
nomline (target-centre chord)	15.0°	13.5°	realistic; 0.5° off the oracle
scint (scint-hit chord)	15.5°	15.0°	surprisingly competitive (long lever arm)

Overall RMS for first : 19.1° (X17), 20.6° (IPC). Bias: positive everywhere (scattering can only open a measured angle on average), +2.5–3.5° for symmetric pairs, +10–20° below min-KE $\approx$ 2 MeV.

Energy dependence (fig 2): σ_{68} vs $\min(KE^-, KE^+)$ falls from $\sim 40^\circ$ at 1 MeV to a plateau of 12.5° (first) / 11.5° (vline) above ~ 8 MeV. The resolution is entirely a soft-track problem; for X17, $KE^- + KE^+ \approx 19.6$ MeV, so one leg is always ≤ 9.8 MeV. (The bias spike at min-KE ≈ 4 MeV in fig 2 is a quantile/ binning artifact of a bimodal residual in the transition region — don't read physics into it.)

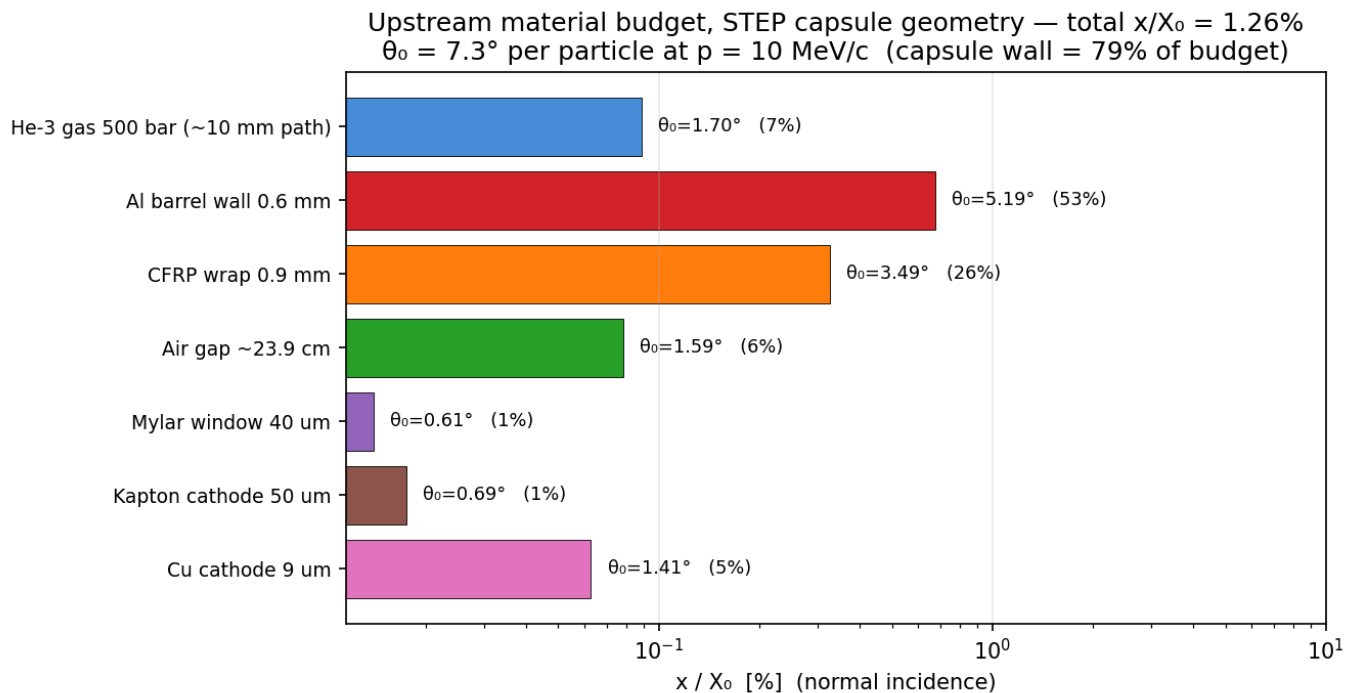


4. Where the resolution is lost: the material budget

Highland-model budget along the radial exit path (normal incidence), new STEP geometry — generated by `scripts/make_angular_resolution_figs.py` :

layer	x/X_0	share	θ_0 @ $p = 10$ MeV/c
He-3 gas, 500 bar, ~10 mm	0.089 %	7 %	1.7°
Al barrel 0.6 mm	0.674 %	53 %	5.2°
CFRP wrap 0.9 mm	0.327 %	26 %	3.5°
air gap ~23.9 cm	0.078 %	6 %	1.6°
Mylar 40 μm + Kapton 50 μm + Cu 9 μm (MM window/cathode)	0.094 %	7 %	$\leq 1.4^\circ$ each
total upstream	1.26 %		7.3°

The measured `first` median (7.9° at $\text{KE} \approx 10$ MeV) sits right on the Highland prediction (median space angle $1.177 \cdot \theta_0 \approx 8.6^\circ$, minus obliquity/tail effects) — the simulation and the analytic budget agree, and the `s_eff` scattering-localization study in `pair_analysis_v2.pdf` independently places the effective scatter point at the capsule wall. **The wall is 79 % of the problem; the 500-bar gas itself is only 7 %.**



5. The vertex constraint: why it is already maxed out

The geometry of the argument: a particle scatters by ψ at the wall, at distance $s \approx 11$ mm from the vertex. The first MM hit is at $L \approx 250$ mm. The chord from the vertex to that hit makes an angle of $\psi \cdot (L-s)/L \approx 0.96 \psi$ with the true initial direction — the scatter is so close to the vertex that the chord inherits essentially all of it. That is the entire story of why `vline` only improves on `first` by $\sim 1^\circ$ (and that 1° mostly comes from the chord averaging over drift-entrance effects, not from "fixing" the wall scatter).

Could a measurement plane *closer* to the target exploit the vertex constraint better? In principle the chord error $\rightarrow 0$ as the plane approaches the scatterer (error $\approx \psi \cdot (1-s/L)$). But the chord needs the vertex, and per-event the vertex is unknown within the capsule: $\sigma \approx 12\text{--}13$ mm along the beam, ≈ 5 mm radially. That adds σ_v/L to the chord error — $\approx 3^\circ$ at $L = 250$ mm, but $11^\circ+$ at $L = 50$ mm. The two requirements (short L to beat the scatter, long L to dilute vertex ignorance) are incompatible; **$L = 250$ mm is already near the optimum for an unknown-vertex chord.**

Can the vertex be measured per event instead? No: the tracks themselves are the only handle, and their median distance-of-closest-approach to the true vertex is 1.0 cm (KE 10–15 MeV) to 3.7 cm (KE 3–4 MeV) — at or above the 1 cm bore radius. The capsule geometry is a *stronger* vertex prior than any two-track fit. This is the quantitative sense in which the small target has already banked the vertex-constraint gain: per particle, assuming the capsule centre costs $\sim 3^\circ$ in quadrature ($9.9^\circ = 9.4^\circ \oplus \sim 3.1^\circ$), and at pair level `nomLine` is 0.5° (X17) from the truth-vertex oracle. **Total remaining headroom from any smarter vertex treatment: $\leq 0.5^\circ$.**

6. Tricks inventory

Already used (in `analyze_pairs.py`):

- direction at first MM hit; PCA full-track fit; hit-position smearing test; edep-weighted centroid chord; true-vertex chord (oracle bound); **target-centre chord (the vertex trick — implemented)**; trigger-scint chord; resolution-vs-min-KE selection study (truth KE).

Available but not yet exercised — worth doing, modest gains:

1. **Bias / response-matrix correction** — the full $P(\theta_{\text{reco}} | \theta_{\text{truth}})$ map exists (`h_open_2d` , and `geant4_response.json` ψ tables). Free.
2. **MM-internal quality cuts as an energy proxy** — straightness χ^2 of the drift track, dE/dx in 30 mm of gas, range. Goal: reject KE $\lesssim 3$ MeV legs and realize the plateau resolution ($\sim 12.5^\circ$) without a calorimeter. This is the one genuinely open reconstruction study.
3. **Per-event vertex-y refinement** from the two first-hit positions ($\leq 0.3^\circ$).
4. **Optimal chord+direction combination** — their errors share the same wall scatter, so the gain is $< 1^\circ$; low priority.

Fundamentally unavailable without calorimetry:

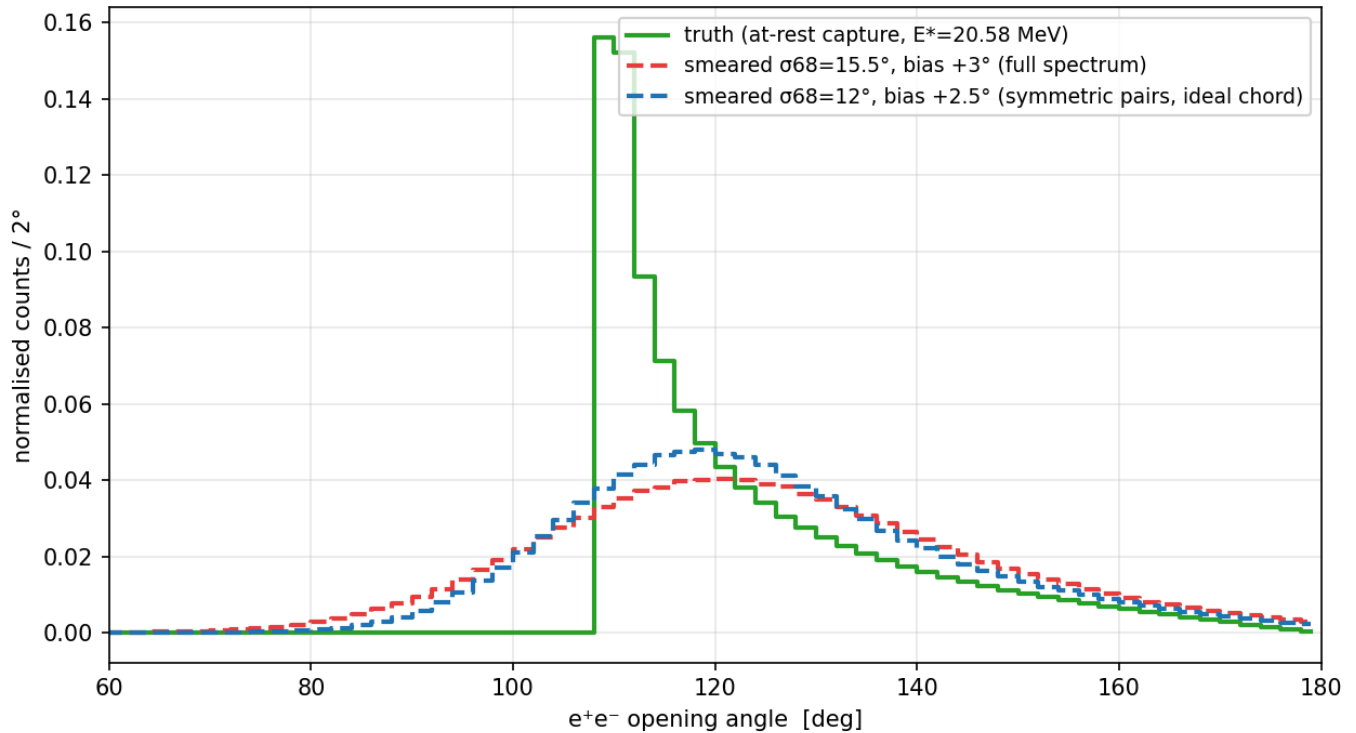
- **Per-event invariant mass.** With only two measured directions there are two unknown momentum magnitudes. Imposing both kinematic constraints of the at-rest capture ($E_1 + E_2 = 20.58$ MeV and $|\vec{p}_1 + \vec{p}_2| = 11.9$ MeV/c) returns $m_{ee} = 16.8$ MeV *identically* — circular. Imposing only one leaves the system underdetermined. The opening angle is the only discriminating observable, and no kinematic fit can sharpen it.

Hardware levers (for completeness — see Part I verdict):

- All-composite vessel (remove the Al barrel, 53 % of budget): $\sigma \times \approx 0.65 \rightarrow \sim 10^\circ$.
- Halving total wall: $\sigma \times \approx 0.8$. Lower gas pressure only helps via a thinner wall and costs capture rate linearly — likely a bad trade.
- Helium bag in the air gap, thinner MM window: ≤ 6 % of budget each, negligible.

7. What the peak looks like, and the recommended path

X17 opening-angle peak dilution by the measured resolution (acceptance not applied; Gaussian-response illustration)



The at-rest truth spectrum has a kinematic shoulder at $\theta_{\min} = 2 \cdot \arcsin(m/E^*) \approx 109^\circ$ with most of the signal within $\sim 20^\circ$ above it. After the measured smearing the shoulder is gone: a broad bump, $\text{FWHM} \approx 40^\circ$, peak amplitude $\times 0.25$. Even the best-hardware scenario ($\sigma_{68} \approx 12^\circ$, blue) looks qualitatively the same. Cut-and-count on θ is dead; what remains is a shape fit:

- signal template: this response matrix $\otimes$ truth spectrum (already in hand);
- background template: IPC opening-angle continuum (smooth, no shoulder) + combinatorial/wall backgrounds from the Stage-4 pile-up sampler;
- statistics: ~ 2.8 double-triggered X17/day in the 0.2–2 MeV window (~ 84 / 30-day run), trigger-level $S/B \approx 0.09$ before angle-region selection.

Recommended next step: wire the smeared templates into a toy likelihood fit (the `MX17_Simulation` fast-MC already loads `geant4_response.json`) and quote expected significance vs run time — that number, not the resolution, decides whether the high-energy measurement survives.

Caveat: all of this is at-rest kinematics. MeV-region captures carry $E^* = 20.58 + \frac{3}{4} E_n$ (up to +1.5 MeV at $E_n = 2$ MeV) and a CM boost along the beam: the truth shoulder shifts/softens by a few degrees and the smeared peak moves accordingly. The resolution itself is material-driven and will not change materially. The generator E_n -kinematics extension (open item) is needed before final template shapes.

Appendix — reproduction

```
# figures in this note (reads analysis/pairs_v2/geant4_response.json):
source ../../PycharmProjects/nToF_x17/.venv/bin/activate
```

```
python scripts/make_angular_resolution_figs.py
```

```
# the underlying 17-section analysis (lxplus, ~100 files):
```

```
python scripts/analyze_pairs.py /eos/experiment/ntof/data/x17/full_sim/pairs_v2_step_target/ \  
-o pair_analysis_v2.pdf --export-response geant4_response.json
```

Numbers quoted from: analysis/pairs_v2/pair_analysis_v2.pdf (method-comparison legends, pointing/DCA section, s_{eff} localization) and analysis/pairs_v2/geant4_response.json (ψ tables, σ_{68} /bias validation block).